The Free Energy Principle and predictive processing

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okay okay good so um can you guys see the full screen yes yeah okay good good all right uh well good to see you all there in Egypt it sounds like a great workshop and um uh you've got some big buddies of mine uh Fernando and Pedro there I think it's somewhere in the background I don't know so you've got a interesting great week ahead with lots of exciting speakers um and I think it's kind of uh an honor for me to be able to present um probably what I would say is one of the most interesting and important um attempts at some sort of a general theory of what's going on in the brain from an algorithmic perspective the Bayesian brain predictive processing active inference and the free energy principle uh so um you know I can see you guys all there um I wonder if if by a show of hands I could like anyone could put up the hand if they feel like uh they they um have a have a grip on the free energy principle and active inference anyone put up their hand how about something easier like anyone like familiar with the kind of basic Bayesian brain hypothesis okay so you know a few so okay we'll we'll okay we'll look we're gonna go at it slowly um and I assume that like most people have a kind of a mathematical background so so that's gonna make it a little easier so we'll start from um philosophy and you know the basic uh the basic place we start is from the nature of perception so you know I'm outside I'm looking at a tree and I kind of just see a tree and it's just amazing but we are thinking well somehow um I form some sort of a representation of that tree in my mind and um and of course what we what we begin to understand is that actually if I've always perceiving just the representation of the tree in my mind then I'm not really perceiving the tree as it is in itself you know that's a sort of philosophical issue um and also the other thing that I must be doing is I'm also forming a representation of myself and that is so uh in our in our minds ah you know we're also forming some sort of representation of ourselves and it's a big issue in philosophy how this works and you know one of the first great philosophers to really uh start to nail this is Emmanuel Kant um uh in the um the 18th century and what he showed is that look um we only experience the phenomenal world the things as they appear to us and we have to when we use the categories of space and time and causality to arrange our perceptual framework and these kind of things like space-time causality are kind of hyper prized that we bring to bear to organize our sensory appearance our sensory data and what he called it is that he thought that he had created some sort of copernican revolution in uh in the in the mind Sciences in the sense that he said look it's objects that must conform to our cognition rather than cognition conforming to objects and this was a huge insight and um you know about a hundred years later helmholtz a very famous physicist system psychologist um started to crystallize this as perception as inference and this is really where um where where the where the free energy principle and um predictive processing starts with when we look out from our eyes unlike what uh sort of Descartes the typical view that Descartes might have helped we are not just passively um experiencing what's coming through our eyes um perception is kind of hard if you look at this video what you see is roughly speaking what we might see in accurately if you could actually see what's coming through your eyes because all that we can perceive through our eyes is a very very small Central area of the of the Immaculate for where um and it's very difficult to see what's going on and by the way you know most of it's in black and white only the central partisan color so if we actually see what I was what you're actually seeing now you can see much clearer that you know you're perceiving this B um buzzing around a flower now it's very difficult to make sense of the the dot on the left but somehow our brain constructs a whole image now how is that done um that is done that is called the Bayesian perception problem and the basic problem is this we receive some sort of that you know that we're assuming that there is something out there in the world which are the causes of our perception actually according to Kant we can't really say much about this at all but what we do get is sensory data so U is some sort of um large Vector of uh sensory data and the problem for the agent is that we have to form something called the recognition density which is the probability probability distribution of like what are the causes of my sensory data given my sensory data what are the causes of my Sensations given my sensory data now that of course is um solved by using um basis theorem um and basis theorem as you as you know this this term here what we're trying to solve for is the posterior the probability of the sensory of the um causes of my Sensations given my sensory data and to solve that we have to use two terms as well as a normalizing term underneath here um we start with our prior probability of what do we before receiving any data what do we expect um the what is our background prior beliefs around the uh causes of this of these features in the world that there are trees in the world for example in their agents and then the second term is the likelihood of the data what is the probability of getting this particular sensory data given that it comes from my best guess which is a tree so in other words How likely am I to to receive these particular Sensations given that there's a tree out there my hypothesis of what's going on and underneath is the probability of receiving this sensory data in the first place which is a kind of normalizing term now that normalizing term actually is is very difficult to compute it's an infinite integral you basically have to go through every single hypothesis of you know this sensory data could be caused by a tree could be caused by a mountain it could be caused by um you know some sort of hallucination you know you couldn't possibly do it it's computation explosive so although Bayes theorem is the mathematically optimal solution that maximizes the value of all information it is absolutely the correct mathematical solution we cannot solve it we cannot solve it using machine learning because it's computationally explosive and evolution cannot solve it our biological beings cannot Solve IT explicitly so how do we get around it well we can draw inspiration from machine learning techniques like artificial neural networks um so artificial neural networks um you know train on on whole loads of of sensory data so whole loads of data for example this particular neuron could train on a whole load of faces and as it learns as we adjust these synaptic weights it starts to do something called feature extraction so at low levels of the sensory data at the low levels of the neural network it extracts low level features like Edge oops sorry edges and corners and little things like this no level features at Mid levels of the network it's extracting mid-level features like noses and eyes and ears and then at high level um areas it extracts the high level features like um whole faces for example but the question is you know how do these networks train these networks um uh train essentially by being provided some ground truth sensory data so what we do is go through something called supervised learning okay so the way supervised Learning Works is that we really know what a true face is you're given all these thousands of examples of true faces and then what happens is that we we get a new face of sensory data and um we we take a guess this network has not yet trained it takes a guess at what this is this could be a cat or a or a dog or a face and what we know is is whatever this um whatever this image is it could be as I said um you know anything a cat or or a dog or a face um we actually know the true category um it could it's actually a face and so we can compare the guess of the neural network with the ground truth the true features and that sets up a prediction error and we use that prediction error signal or generally we Square it uh uh because then the derivative is very simple and we we basically you know apply multivariate calculus to minimize the prediction error and adjust all these synaptic weights so that the error is is minimized and and step by step the the neural the neurons start learning that's a very famous algorithm called the back propagation algorithm now the problem with that is that there is no supervisor in the brain who is giving us the ground truth data who is saying this is a cat this is a dog this is a face there isn't anyone even if you're told as a child you know this is a cat two or three times that's simply not enough data to kind of do uh supervised learning so somehow the brain must be doing this in an unsupervised way so how might we be doing that well there's only one thing we know okay we don't know the ground truth the only one thing we know is the data itself and so if we have to guess if we only know the data itself and we're trying to guess a feature Vector in other words what are the features of my experience dogs cats faces um edges you know I mean whatever features there are what we can do is a brilliant strategy which is to try to uh take a get try to learn this net feature Network and then whatever we guess at we then say let me then assume that that's the feature in the of the world those are the causes of my sensory data and let me generate a network going the other way to construct some predicted some predicted data let me simulate what I would be seeing if there was a cat and a dog and a face in this picture and that is um and and then we can compare the predicted data that I the the predict what what I'm what I'm guessing are the predictions and the actual sensory data and that's that's a prediction error and that is used to train the network now this kind of network is called um in general an autoencoder and uh there are three parts of the network the first part is the recognition model or the feed forward model uh which is which is trying to go from the data to a feature Vector the the code this gray bar here is called the code if any of you have heard of predictive coding well the code that they're referring to in predictive coding is basically this line here this series of neurons in the middle here and then the second part is the generative model this is generating what I would be seeing um if this was the correct uh features of the world and that second part of the model is called a decoder as well so the problem is that this kind of network is extremely difficult to train um because it's very non-linear it's very you know there are too many possibilities and and data is ambiguous and difficult to decipher um these kind of networks especially the type of network that called the variational autoencoders are trained using um something called free energy as an objective function rather than just the prediction error and free energy takes into account the prediction error that we talked about Plus it takes into account prize prior probability around what you would expect this feature Vector to be all right um and that gives us a a principled way to train the network so this kind of network that we just showed in with Gene learning we can take inspiration from that inspiration and last week I was with uh all week with a whole lot of scientists including um uh Carl friston who's who's really one of the the founders of the free energy principle in fact the the kind of key architect and he actually warned us about um kind of trying to use machine um analogies however he felt that this particular analogy is a very good one and so I commended to you to really think about how the brain is working so we we're using this model as a kind of idea of how the brain is working except that in the brain what happens is that this top Network can you this this green Network this generative model actually folds back onto the feed forward Network so I don't know if this animation is coming through clearly but that was supposed to illustrate the network folding back onto itself and the feed forward Network the blue Network kind of goes up and in fact what actually travels up according to the the predictive processing framework is not the data itself but simply the prediction errors simply these prediction errors travel up and what's actually happening is that this generative model is taking a guess from the latent layer the code and the feature vector and generating what I would be seeing and and the only thing that's traveling up is the prediction error as compared to this charitable model so this is the fundamental idea of what is going on in predictor pressing except that this is probably happening in a hierarchical way we have a number of these kind of models that are going on organized in sort of layers and and then what happens is that these criers are actually arising from higher level models in the brain um and and prediction errors actually go up to these higher level models and prize for this higher level model are coming from even higher level models in the brain and so they could be maybe you know who knows how many layers in this you know perhaps half a dozen layers in this kind of hierarchical predictive processing framework so let's look at specifically how the variational algorithm is supposed to work now as we said we we see you know there are certain features in the world that uh we're assuming are generating our sensory data this sensory data these features in the world are generating our and we cannot solve bayes's theorem correctly so what we have to do is we have to form an inference we have to make a guess at an approach approximate posterior so we make a functional form often you know in these kind of models they're often you assume to be gaussian for example you know multi-dimensional gaussian um with certain parameters so this probability distribution have certain parameters Phi and that is instantiated in the example that I gave it's operationalized in the feed forward Network that we talked about the recognition model the blue Network at the bottom of the auto encoder and from this approximate posterior we then have an estimate of the code this this red bar here which is the features an estimate of the features in the world that there's a tree that there's an agent and from there we then create we have a generative model which in the brain of course feeds back um on itself and the output of the generative model is a is a the sensory data that I would be my expectations of what I would be seeing in the world in this case I don't know if you can see this this sort of little tree that sits on top of the generator model and from there we then are in a position to compare the predicted data U Dash and the actual data and we get a prediction error and that is at the heart of trying to train this model but as I said before there's one more factor that goes into the free energy and that is the deviation from my prior expectations how different are these features from my prior expectation so for example um if it if just using the prediction you know if I thought this was a alien spacecraft um that had landed that would have a lot kind of a low prior probability and so uh that would mean that these causes are kind of far away from my prior and um that would mean that I would be penalized for thinking it's a alien and using this free energy which combines both the prediction error and this kale Divergence which is essentially the distance or some sort of inverted commas distance it's not actually a metric um from my prior expectations I can then train this this recognition model and the generator model itself to minimize free energy and I will hopefully slowly slowly get a convergence between this approximate posterior and the real posterior you know a really accurate image of what's going on and that in a nutshell is the variational algorithm to solve approximate Bayesian inference now uh before we move on um again uh show of hands I just I I know you're sort of a mathematical crew but how many people are familiar with a relative entropy KL Divergence can someone can you have a show of hands please okay I can see maybe two three okay fine so this is a this is really important and and um since I understand you're supposed to be like understanding the stuff at a technical basis so let me let me let me just go through this because this is kind of important it's very easy I'm not gonna um but but this this basic idea is that the KL Divergence between two probability distributions is essentially measuring something like a distance that you know how far away are these distributions how different are these distributions and the particular form that that this functional form has is is you basically compute you take uh this the first probability distribution uh P log of p over Q okay and so these are the two distributions you're trying to compare but what it really means the take-home message that I want you to say is what it really means is how much information how much information suppose that I thought that um a Treasurer was buried on a straight line let's say a road going between zero and one okay and I have no information otherwise my my kind of probability distribution is um Q of X I don't know where the treasurer is buried it could be anywhere along this line it's me a kind of a more accurate map saying actually the treasures buried between 0.87825 0.825 and one right here P of X this is a new information that I get a more accurate probability is Fusion how much information have I got how many bits do I get from going from Q of x to P of X what is the informational distance and that uh informational distance is roughly speaking how many yes no questions I have how many bits of information how many yes no questions I have to go from Q of x to P of X and you know it's kind of easy I could say is it on the left of Q of X or the right of Q of X that's question one so it's on the right okay if it's on the right is it on the first half or the second half it's on the second half okay that's the second question is it then I say is it on the first half of the second half of that and it's in the second half so it's p of X so three questions so in fact if I compute this this uh sum and let's just do it like an integral because it's the same thing what I see is um that I can actually compute that it's pretty easy integrals like this this is eight and then that's log of 8 p of X is is a height of eight because the area under the curve is one and Q of X is one and times one you know the error on the curve and if I compute that I get exactly three bits because this this this integral limits give gives me 1 8. okay so anyway just wanted to explain that um that it's a very simple calculation it's roughly speaking the informational distance I I worry about using the word distance and Metric if you want a real metric you have to use something called the Fischer information distance and you integrate this KL Divergence and then you get you know you get into the kind of wonderful informational geometry that I'm sure Fernando is going to be talking to you about right as I said this is not a metric because it's not symmetric but let's not worry about that now let's get into the real heart of it because you're here to learn about the free energy principle and I'm only going through this not to kind of uh do a shock in all but actually to say in fact how simple this is you will see as you go through your you know as you go through uh your careers in this whole area that people are Bamboozled by all the mathematics around the free energy principle and actually it's relatively easy to to do but um it's couched in very technical uh uh tensor calculus okay and it's kind of difficult to unravel so I hope you will later on come back and say oh well that was a beautiful explanation that you saw here in Egypt so the basic idea is is um to solve the fundamental problem that we talked about what are the causes of my sensory data and we're going to use basis theorem but as we just saw we cannot use basic serum directly because it's computation explosive so we're going to use a variational algorithm we're going to approximate the actual posterior that we're trying to learn with we're going to approximate it with this Q of this Q distribution which is you know for example could be a simple form like a gaussian with some parameters multi-guard um uh distribution um we are going to uh try to minimize the distance inverted commas um technically the KL Divergence between q and P we're going to minimize the Callback Library Divergence between the Q distribution and the true posterior P by changing the parameters of the underlying distribute distribution okay so what does that mean uh does this come across this this equation can everyone see it yeah yeah okay good so this says please minimize um by changing these parameters of the Q distribution the KL Divergence between p and Q all right how do we do that well um I showed you the functional form of what a KL Divergence looks like this is just the definition of it all right that sounds straightforward enough and we just sum over all the categories V all the particular features it could be in this now you'll immediately recognize that when I sum with the probability in front that's basically an expectation value right so I can basically rewrite that as the expectation of this log of Q over P um so here's the expectation under the Q measure this is a measure of probability measure under the Q probability measure it's it's this distance now of course I've just put a minus sign instead of dividing because that's just a property of logs okay you can separate out the logs in this way very simple and then we're just going to use bases theorem because what we recognize is look this is the probability of of the features given my data and that is here on the side that is just the posterior well if I take logs of this posterior then I get a log in front on this side and of course when I multiply um things the logs add and when I divide the the logs uh you subtract so essentially this here which is the positive the law the the log of the posterior simply becomes this which in these these are just the terms the likelihood of the data that's the term as this term here this is the prior the log of the prior and this is the log of the probability of the data the normalizing term okay so this is just applying basis theorem and then we just rewrite this in very simple terms we simply take the prior and put it in with the first term and then we separate out the other two terms okay and what we then have a simple rewriting is we recognize that the first term here is nothing but another KL Divergence of this form except that what we have is a KL Divergence between the approximate posterior this this this this um this Q distribution and the prior probability of V okay and then these terms here are simply the expected value under the Q uh measure of the um log probability of the sensory data of the likelihood of the data and then this term here this term here I can take out of the by the way I could take out of the expectation operator because it doesn't depend on uh Q at all right so there's you know there's nothing stochastic here so I can just take it out and then finally so we can notice that it's a constant now these first two terms are the free energy um this first term here um the KL Divergence and then the second term the um the expected uh surprise I'm gonna come on to see why that's called surprise so this term here is roughly speaking how surprising is the sensory data what is the expected surprise of the scent of this data given that

I'm assuming that given this feature given that I think it's a tree how surprising is this sensory data so if you if you look it's a negative log if you remember logs go from minus infinity to zero so if we put a negative sign it goes from plus infinity all the way down to to zero and um if you have um if the if it's very very surprising if it's not very surprising at all in other words this is precisely the kind of sensory data that I would expect given that it's a tree it's not very surprising um it's got a it's got a low it's got a high probability of being this kind of sensory data given that it's a tree then it's not very surprising if on on the other hand it's like really unsurprising you know I'm seeing something that is more like um uh snakes and crocodiles then um it's very surprising data and so this term would be highly surprising so the first term in the in this term in the free energy is surprise how surprising is my sensory data in other words how good a fit is this explanation for the data the second term on this the first leading term here is the KL Divergence between the um between the uh the the posterior that I've that I've learned and the prior so take a look at this furry animal here it could be actually let's suppose that it's equally likely to be a dog a cute little furry dog with blonde hair or a polar bear okay in other words let's suppose that the surprise of the sensory data given that it's a polar bear is equal to the surprise that it's that it's a dog in other words they in other words the data fits both the dog model and the polar bear model but polar bears especially walking around in Egypt are very unlikely the prior probability of seeing a polar bear in Egypt is very low and so this term would be a very high distance between the posterior that it's a polar bear and uh and the prior that's better so you'd be very much penalized remember we're trying to minimize free energy here okay so these are the two terms of of free energy now I'm sure you're probably asking and no one ever explains in in in Neuroscience why is this called free energy I mean I I thought the free energy was something to do with entropy and energy and like stuff like that you know Jewels you know this is information this is not Jules well it's called the free energy because you can rewrite this term I won't explain how this is very easy to rewrite this from from this line here into a slightly different format and you can check it for yourself that it's possible to do this and this happens to be this way of rewriting this free energy term here happens to be equal to exact I mean isomorphic with the helmholtz free energy in thermodynamics okay and the reason for that is because this term here is the average energy um you remember you when you sum up and multiply by probability uh this this thing here which which is going to turn out to be the energy of the explanation that's the average energy and this is the entropy minus log P log uh uh uh uh P log P sum is the entropy okay so here's the end so energy minus entropy or average energy minus entropy if as you might recognize is the typical form of a free energy and and then you might say well why on Earth is this an energy this is an energy if you you know for those you remember you know have done physics the boltzmann distribution in thermodynamics is that the probability is related to the energy by an exponential term so if I take log of both sides I get rid of this exponential term so then the minus sign comes over and hay pressure minus log p and that is equal to the energy so that's called the energy of the explanation the energy of the explanation that this is a tree so that is the free energy so there you have it and now you can forget all about it because you've seen it once in your life once you've seen it in your life it's been demystified and you can say uh it wasn't magic after all um I I need to move on a little bit faster so I'm actually just going to to whiz through some of these uh bits but but roughly seeing that um what this is basically saying is that is that um there's a very nice approximation and and of this free energy principle and the takeaway message is that the posterior belief the posterior that you're forming is roughly speaking the weighted average of your priors and your sensory evidence your priors and the sensory evidence coming from the likelihood and the the weights of that weighted average turn out to be the precisions of the distribution in other words how if your Precision is inverse variance in other words if your prior is particularly sharply focused you know low standard deviation then it's going to be highly weighted within the posterior or if you get very accurate data okay you're gonna um highly weight it within your posterior okay um right so so we've got this basic hierarchical model and these precisions that we talked about how sharply influenced are going to influence the in the top-down ratings top down weighting of the priors compared to the sensory data now in the brain we've got a kind of cortical hierarchy and that is the the hierarchy that I'm talking about in the in the um hierarchical predictive processing framework is precisely going to be that um uh it's it's a homologue of the hierarchy in in the cortex and this is a very nice machine learning um exercise that was done to try to figure out the cortical hierarchy and you know very similar similar things to to what one might imagine you know low level sensory percept so low level features mid-level features like motion and rotation and faces and then at the right at the top of the cortical hierarchy are language and Concepts and reasoning as you might expect and of course this hierarchy is probably best not represented as a as a pyramid structure but in fact from a hierarchical from a predictive pressing point of view it's really seen as these these different channels of um streams of hierarchical predictive pricing models until they start integrating together perhaps in some sort of a global workspace or something that you might learn about of course no one has put together predictive processing and Global workspace Theory as yet people have tried to attempt it to do that but I'm just kind of warming you up and making you salivate to say that um you know maybe these things could fit together in some way you know you have these uh predictive pressing models of vision audition the motor system actually goes outwards olfaction somatosensory interception gas station the whole visceral proprioception um and they kind of come together now actually we're probably not going to have time to go through the phenomenal self model but it's super interesting that in fact the self itself is nothing but an internal model yeah within a predictive pressing hierarchy uh hierarchical framework so the basic upshot is that our conscious experience to the extent that we're conscious of the generative model will be the contents of our general model that is the take-home message at what will become a uh kind of predictive processing Theory Of Consciousness as yet it hasn't been worked out but if if it is worked out um the new the computational neural phenomenology assumption is that the contents of our generative model is the contents of our of our conscious experience to the extent that we're we're conscious of it so this is not completely worked out because we've got to figure out what are we conscious of but in other words you know if I'm looking out of the tree I'm going to have like a represent you know a you know in my generative model this this construction of a tree and a and an avatar you know the self Avatar and some thoughts and these thoughts could be visual thought course and and auditory thoughts blah blah blah and visceral Sensations and proprioception um and smells and sensations of the breath at my nose and my the sense of my heartbeat and sounds and emotions I can like you know I'm constructing my emotions in in a sense like uh as as Lisa film and Barrett is kind of described in this way and all these contents of the generative model aren't just sitting on top of each other until unless you're uh kind of doing psychedelics or in a meditative state they're organized into a phenomenal self model so what does that look like well they are reorganized I don't know if that that animation came through okay I'll do it again they are organized into a subject object Duality so everything in this blue area here is within a subject uh there are internal interception these are extra reception um there's a certain perspectivity there's a subject object relationship there's a notion of agency there's multi-sensory integration um there's a sense of minus a kind of a sense of a Cartesian theater these are all constructions and that we're not really aware of the fact that there is this phenomenal self model and that's called phenomenal transparency according to the philosopher Thomas metzinger so just to before I finish just to to finish to finish up because the the real fruit of this whole free energy principle and pretty depressing framework is something absolutely um amazing which is that we can put action into the same framework to minimize free energy and that is called active inference and to be honest I think that that's a real that's though that's why this framework is so amazing because it kind of unifies action and perception in a single framework and Carl friston came up with with this kind of idea in the in the um maybe around 2008 and nine um and and the way it works is that essentially you the way you could put it is that the goal States let's say I'm trying to reach over to grab an apple um uh you know and the apple is here in green if you can see and I'm currently here well if the way it sets up is it puts that getting the apple is in my prior I have a prior expectation to get the Apple that's kind of my reward goal states are embedded into the price and then immediately there's a prediction error between where my finger is at the moment my hand is at the moment and where I want it to be getting the apple right and so this prediction error will be resolved not by changing my perception but by moving my muscles in other words doing action um and so inference and action are resolved together and there is a movement trajectory until the prediction error is minimized and I got the Apple absolutely astounding so the way okay I won't go through this but this is this is what you're gonna finally understand later on and I just don't want you to be mesmerized by this but this is um the kind of Final Act of inference framework with action in it and it's typically instantiated in a partially observed Markov decision-making process under active inference so the way this works is that what we saw before is that I'm trying to predict my um my sensory data these o is the outcomes the outcomes of my kind of you know the the World shows me my sensory data these this is this is the sensory data and I'm trying to model the world and that's these states these are internal models of the world and these matrices uh which which a probability transition matrices don't worry about the details about these you know how these are formed what I need to do when I bring action in is that I need to have some sort of preferences some sort of goal stage which are called preferred um preferred observations this is this Pro this is a probability distribution over my preferred outcomes and that is gives me my my expected free energy and uh the expected free energy and then I have my prize around my habits uh you know your habitual actions and the Precision waiting over this expected free energy and that gives me my policies of the world and the whole thing is minimizing expected free energy that is a uh that's a quick Whirlwind Tour on um on this framework and we can go to questions now I can hear us yes we can right okay good when you see me as well yeah Okay cool so who's good questions okay let me saw you first there you go [Music] uh um the um machine learning analogy that you side reflect it like seems to uh suggests that your signal kind of enter and learning happening in the brain or like um like some like other aspects of the maybe yours can you hold it because can you hear me well it goes rather as if you're talking underwater is this okay yeah yeah uh start over the machine learning analogy your side reflects seem to suggest that your singer for Android learning will be happening in the brain where like um like a from like other aspects of like free energy principle that I have read about it's um organization uh I was wondering very few could comment on if that can be brought together in some sense or if that's like or that does the contradictory views okay yeah look I think I think you have I think you have uh put your finger on a very important uh issue which is that you know I don't think we have the granu well first of all take the machine learning phenology as simply pedagogical right uh uh this is really to introduce and try to make sense of the pre-energy um how is being instantiating in the brain is something we we really don't have um good models and and data over yet uh so I think it's a great question and I think you should think about working on [Laughter] other questions uh I think that is first um I think you can ask the question okay hi Shamil all right good to see you here I didn't I didn't know you're out there so great can you hear me yeah okay great um so uh Predator processing is not in itself a theory of Consciousness uh but clearly it has implications for Consciousness I noticed that you framed most of the discussion in terms of perception rather than Consciousness and it seems like free energy principle in general comments a lot more explicitly on that rather than Consciousness um does Carl or anybody else um who is working on the free energy um explicitly distinguished between perception and Consciousness what is the difference right can they be treated as the same okay so as I mentioned the whole of last week I was at a workshop with Carl friston Anil Seth Jacob Howie um Thomas metzing you know the you know the whole leaders of this whole area um on computational neural phenomenology and that what you're asking was precisely what we were debating back and forth here you know and it was very it was uh very interesting so I would say that so you know we are right at the beginning of of saying look we have our conscious experience and we know that the and we have the neural going on and we know that the brain must be solving some sort of Bayesian influence problem so it's a good way of approximating what's going on in the brain is by modeling it as kind of a predictive pressing framework and given that we are trying to given that a phenomenal experience is you know if you take a functionalist perspective a phenomenal experience is somehow being worked out in terms of neural goings-on um you can then say that the predictive processing model is going to be a homologue of what is going on in the brain but also what is going on in phenomenology so we're precisely trying to draw you're absolutely right to to to to to to ask this question but we are explicitly say trying to draw an analogy to say yes indeed this is a way the the project of of computational neural phenomenology is precisely um active inference Frameworks for example or other computational models to model our phenomenal experience and and if you want to drill down further if you're assuming some sort of active inference framework then I would even say that that um parts of the posterior or the posterior the homologue of what is our conscious experience and our Precision weightings are the homologue of our attention mechanism in other words the sort of you know if I if I really pay attention to um this bottle of water then in my posterior remember the Precision waiting is the is is is is the weight that that's used as the to calculate the weighted average in my posterior so you know this is strongly weighted into my into my posterior distribution otherwise my my Consciousness is deeply in dominated by this bottle okay so great question you're right at The Cutting Edge we don't have a theory of Consciousness which is based around predictive pressing however what I'm trying to say to you is that it's starting to look very evocative thank you so much any other questions yep so actually my question is exactly pretty similar to Ken's question so actually I I've read from a lot of places like oh people always supporting processing active reference is not really a theory of Consciousness it's just a framework but at the same time I got so confused so for example I see a lot of adversary anniversarial collaboration is now trying to test the prediction of predated processing or active inference between these predictivity theories and other theories so it's a framework or it's a theory if it's just a framework then it does not really review for example necessary review for example in the background masking why sometimes I see the stimuli and why the other times I do not yeah and and and if this if it is a theory then it is a theory then it's normal framework then it should give an explanation to so for example why backward masking binocular rivalry works so guys you know guys and girls what you're seeing here is how science is actually you know the the sausage making there is a you know there are people Anil Seth and Jacob Howie have written a paper to say predictive processing is not a theory of consciousness and then other people like uh Adam saffron a couple are just a year you know go kind of created something like a really interesting Theory Of Consciousness called integrated World model Theory which is essentially integrating aspects of Predator processing Theory and combining it with the global workspace theory in the kind of sense that I alluded to about sharing of information at the center of the ring and actually also combining it with areas of the back of the brain in IIT terms right and so actually integrating it in and basically what you're seeing the kind of maneuvering in the field where people think ah we're kind of grappling towards a predictive pressing Theory Of Consciousness it's not yet clear it's not like 20 years old like uh or 20 30 years old like Global workspace Theory or iot or something you know it's we're right at the beginnings of this I predict that we will have a you know a theory of Consciousness based around predictive processing I also predict that people will say that's not a theory just like people say other theories in our Theory you know and there'll be lots of argument and there'll be you know and I think that will be a richer process for thinking about Consciousness because actually I'm kind of partial to the view that recurrent processing theories IIT theories Global workspace theories higher order theories and predictive processing theories are all bringing something important to the table and actually at a at assc which is the main Consciousness conference last year um there was a debate in that whole thing between these four theories and there was supposed to be a fifth Theory except that Anil Seth had long covered so he couldn't come but you know there were five theories that were put up for debate so good question but you know see the messy process of Science in front of you yeah yeah yeah I I'm just really confused it's like for example even a new says like at the Wayne Palm paper he just say one thing oh is it pretty it's just a framework but at the same time you can also see his reviews in nature reviews uh Neuroscience lately like a serious Consciousness is still listed as one other Theory and he is in participate but if you know that is not the theory why do you still participate in anniversary collaboration testing it as a theory of Consciousness I just you know I just don't get it also like it's it's it's it's it's wonderful it's wonderful and nuanced and in that Nuance we might get progress unfortunately at The Cutting Edge science is always confusing you know literally everyone at the conference lasts a week was confused and that was a sign of progress thank you anymore you know I think the teachers should feel free to ask questions as well uh yeah hi thank you for your talk um I was just wondering because actually the free energy principle also tries to explain more than just the brain right um so it's more or less a theory of everything so how is your take on that like what can predictive processing tell us more than about the brain okay great yeah that's a very good point and I didn't bring that out but thank you for asking a question on that confrustin often says that there are two ways into the free energy principle one is the low road and one is the high road and I took the low road which is through um inference and then there's another road which is called the high road which based around the concept of Markov blankets and Markov blankets are essentially think of them like um like informational boundaries like for example a cell membrane for a cell that all information has to go through the cell membrane and so if you look at the informational states of the cell membrane you will get that all the information inside will that has come from outside will have to have been represented on the on the blanket states on the bat on the mark of blanket

States it's it's essentially saying that if you have a Markov blanket then you are going to be instantiating a model of the external data and this is very powerful and this you know the the sort of principled information theoretic arguments that have been proofs that have been trying to be given for this and this means that everywhere you see a Markov blanket not just in the brain but in the body perhaps in the organs perhaps in the cells perhaps ecosystems perhaps the planet Earth you know I mean even species I mean you know even you know within in the in the evolutionary tree and species and things like that you know you're going to see this kind of what um is called in the in the area Bayesian mechanics um and I think it's a wonderful idea it's a very powerful um idea you know and um it's very evocative so yeah just that's a brief Glimpse um of of how that is and I've just focused on the brain yeah but there is uh is there a way to detect a mark of blanket like when you see something like okay this is a mark of blanket and this is not one or is that just like you could basically see it everywhere depending on your definition or yeah no that's that's a very good question to be honest I think most markup blankets are only approximate and and that's an issue um in theory but but you basically have to see so the canonical example would be some sort of system that is Auto poetic and self-organizing and and continuing through time so if a system exists so this is how the I'm sketching a little bit how the proof works if a system exists the nature of existence is that it's a system for itself which means that there is a Markov blanket so just to call it that there exists a system instead of saying that we can define a system self-contained kind of system like this which means that the fact that it's self-contained in this way means that it's got a Markov blanket so this is a great question because um because you know this stuff is very powerful I mean there was a paper written a couple of years ago which was saying a general theory of everything every space thing and like officer like the thing was things that exist I mean it was really deep stuff and anything that exists exists with itself has a Markov blanket because that's what it means that it persists through time and one of the just just just very quickly because it's kind of a cool thing if it persists through time as a system one of the things that it must be doing is also in practice not being broken down by entropy surviving against the SEC the threat of the second law uh uh unless it's thermodynamic entropy and so a system has a model of its external environment you know and and acts in such a way as to avoid the second law so this is a sort of interesting argument to say that things like homeostasis and action are are ways of instantiating active inference um in order to or as a result they it means the system continues to exist which is resists the second North thermodynamics very interesting thank you so now that we have applied free energy to every space thing uh I think we can finish here thank you very much Emil for the amazing talk and uh yep all right thank you so much thank you and good luck for the rest of this Workshop